
ALPACA HACKATHON 2026 SUBMISSION

AEGIS // PRIVATE WEALTH

Autonomous Adaptive Portfolio Hedge Agent

A closed-loop multi-agent risk engine that transforms live Alpaca portfolios into self-protecting assets. Evaluates competing options hedges, enforces deterministic risk gates, executes multi-leg orders, and dynamically rebalances as market regimes shift.

MULTI-AGENT REASONING

LangGraph DAG + LLM Tournament

DETERMINISTIC RISK GATE

Zero Hallucinations • Hard Quant Math

CLOSED-LOOP ADAPTATION

Level-1 Drift & Level-2 De-Hedging

Traditional Portfolio Hedging is Broken

Why static, manual, and heuristic-based hedging strategies consistently destroy portfolio alpha.

01

Naive Heuristics & Blind Rules

Common industry rules like 'If down 10%, buy puts' fail completely in real markets.

- Ignores market regime and why the asset is dropping
- Blind to implied volatility (IV) inflation and skew
- Buys overpriced protection at the market bottom
- Zero awareness of sector basis risk vs systemic shock

02

Excessive Carry Cost & Alpha Drag

Passive long puts act as a relentless tax on long-term compound performance.

- Bleeds 5% to 9% annualized alpha through theta decay
- Unfunded single-leg puts suffer severe time decay
- Fails to leverage spreads or collars to offset premium
- Investors abandon protection just before crises occur

03

The Unsolved 'Exit Problem'

Traditional managers know when to buy protection, but never when to dismantle it.

- Hedges are left on as markets recover, killing upside
- Manual adjustment is lagged, emotional, and slow
- Legging risk: partial fills during multi-leg rebalancing
- No automated feedback loop to harvest option profits

CORE TAKEAWAY: Portfolios do not need static rules or speculative predictions – they need continuous, adaptive risk management that actively dials protection up AND down.

The Solution: Continuous Closed-Loop Defense

AEGIS operates as an autonomous risk engine that continuously answers 5 foundational questions.

Q1

Risk Detection

What specific risks exist?

Audits portfolio beta, factor concentration, drawdown, and macro market volatility shocks in real time.

Q2

Protection Sizing

How much protection is needed?

Calculates exact mathematical hedge ratios using Black-Scholes Greeks, Cornish-Fisher VaR, and budget limits.

Q3

Option Discovery

Which structures are viable?

Screens Alpaca option chains across strike, expiration, liquidity, IV skew, and multi-leg feasibility.

Q4

Strategy Tournament

Which trade-off is optimal?

Compares 4 competing strategy hypotheses across 8 quantitative dimensions to select the best fit.

Q5

Adaptive Lifecycle

Should we adjust or remove?

Continuously monitors hedge drift and market stabilization to increase, reduce, roll, or dismantle protection.

THE NORTH STAR: We do not predict market tops or bottoms. We continuously measure risk and dynamically adapt protection.

Agents Reason. Tools Provide. Deterministic Code Enforces.

Strict separation of concerns: LLMs provide qualitative judgment; pure Python math provides absolute safety.

AGENTS REASON

LLM Contextual Reasoning Layer

- Analyzes macroeconomic market regimes
- Evaluates qualitative trade-offs between hedge candidates
- Synthesizes breaking news sentiment & shock events
- Formulates strategic hypotheses with plain-English rationales
- Dual-provider: Featherless AI (Submission) & OpenRouter (Dev)

TOOLS PROVIDE

Integration & Ingestion Layer

- Alpaca Trading & Market Data API streaming
- Full option chain retrieval & liquidity filtering
- Real-time news ingestion and corporate actions
- Alpaca MCP server interface for tool calling
- Persistent PostgreSQL state & auditable agent run logs

DETERMINISTIC CODE ENFORCES

Quantitative Safety & Risk Gate

- Zero hallucinated math — LLMs never compute Greeks or VaR
- Black-Scholes Delta (Δ), Gamma (Γ), Theta (Θ), Vega (v)
- Parametric & Cornish-Fisher Value-at-Risk (VaR)
- Hard risk limits: maximum budget, notional, and position size
- Non-negotiable veto: Risk Agent overrides any LLM decision

MANDATORY SAFETY RULE: If an LLM proposes a trade that breaches the maximum hedge budget by \$1, the deterministic Risk Gate instantly vetos or resizes the order without exception.

Stateful LangGraph DAG & The HedgeContext

Eliminating context dilution through specialized agent responsibilities and standardized data contracts.

THE 5 SPECIALIZED INGESTION AGENTS

► Portfolio Management Agent

Audits holdings, liquidation value, concentration, beta, and current hedge delta.

► Stock Analysis Agent

Computes idiosyncratic momentum, realized volatility, and correlation matrices.

► Market Analysis Agent

Tracks SPY/QQQ technicals, macro regimes, and VIX term structure.

► News Analysis Agent

Filters high-impact earnings releases, macro catalysts, and tail-risk headlines.

► Options Analysis Agent

Parses Alpaca option chains, filtering by moneyness, open interest, and bid-ask spreads.

THE STANDARDIZED HEDGECONTEXT CONTRACT

All raw telemetry is assembled into a single immutable, validated Pydantic contract. Downstream strategy agents receive clean, pre-computed inputs to prevent token bloat and hallucination:

- `portfolio_state`: AUM, cash, positions, beta, drawdown, existing hedges
- `market_state`: Regime (Bull/Bear/Shock), VIX level, trend strength
- `stock_state`: Idiosyncratic volatility, contribution to portfolio VaR
- `news_context`: High-priority risk events, sentiment impact score
- `option_candidates`: Pre-filtered liquid strikes with verified Black-Scholes Greeks

WORKFLOW: Ingestion Agents → HedgeContext → 4 Strategy Agents → Strategy Manager → Risk Gate → Alpaca Fill

The Strategy Arena: 4 Competing Option Architectures

Specialized agents formulate competing hypotheses evaluated across 8 quantitative dimensions.

PROTECTIVE PUT

Direct Downside Floor

Structure: Buy OTM Put

+ Advantage: Full catastrophic protection with 100% upside retention.

- Trade-off: High cash premium, negative theta carry bleed.

► Regime: High-conviction crash or imminent shock regime.

BEAR PUT SPREAD

Corridor Protection

Structure: Buy High Put + Sell Low Put

+ Advantage: Substantially reduced net premium cost with defined protection zone.

- Trade-off: Protection capped below the short put strike.

► Regime: Moderate drawdown defense within hedge budget.

COLLAR STRUCTURE

Self-Funded Hedge

Structure: Buy OTM Put + Sell OTM Call

+ Advantage: Near-zero cash outlay; short call fully funds the long put.

- Trade-off: Upside potential strictly capped at short call strike.

► Regime: High IV environments where cash protection is overpriced.

NO-HEDGE DISCIPLINE

Capital Preservation

Structure: No Trade / Maintain Cash

+ Advantage: Zero premium spend; 100% upside participation and liquidity.

- Trade-off: Exposed to unhedged market tail risk.

► Regime: Low volatility, healthy portfolio, or unfavorable option skew.

STRATEGY MANAGER 8-D CRITERIA: Basis Risk • Net Carry Cost • Greeks Profile • Payoff Convexity • IV Surface Fit • Liquidity • Sizing • Opportunity Cost

Deterministic Risk Gate & Alpaca Execution

No trade touches Alpaca paper brokerage without clearing 6 deterministic mathematical guardrails.

THE 6 HARD DETERMINISTIC RISK GUARDRAILS

✓ 1. Max Hedge Ratio Cap

Enforces maximum 40% portfolio notional protection to prevent over-hedging.

✓ 2. Cash Budget Ceiling

Hard cap on option premium spend (e.g. max 2.5% of total portfolio value).

✓ 3. Liquidity Screening

Rejects contracts with wide bid-ask spreads or open interest below threshold.

✓ 4. Tenor Safety Bounds

Filters out ultra-short theta death (<7 DTE) and illiquid LEAPS (>90 DTE).

✓ 5. Greeks Sensitivity Cap

Limits portfolio Delta drift (Δ) and excessive Gamma exposure (Γ).

✓ 6. Margin & Buying Power

Pre-flight check verifying account cash and buying power sufficiency.

NATIVE ALPACA MULTI-LEG BROKERAGE EXECUTION

► Atomic Multi-Leg Orders

Executes Put Spreads and Collars natively via Alpaca API to completely eliminate 'legging risk' (partial single-leg fills).

► Alpaca MCP Integration

Standardized Model Context Protocol tool endpoints for portfolio queries, option chain discovery, and order lifecycle.

► Slippage & Fill Tracking

Measures fill prices against midpoint and limits, logging exact execution slippage to database.

► Auditable State Transition

Fills update portfolio snapshots and hedge state; fails trigger automatic recovery workflows.

FAIL-SAFE ASSURANCE: The Execution Agent has zero discretion to modify approved orders. It only executes what the Risk Gate explicitly approved.

The Adaptive Lifecycle: 2-Tier Monitoring Loop

How AEGIS prevents overtrading while actively dismantling protection when market danger subsides.

TIER-1: TICK-LEVEL DRIFT DETECTION

► Hedge Drift Tracking

Calculates the spread between Current Hedge Ratio and Target Hedge Ratio in real time.

► Statistical Deadbands ($\pm 2.5\%$)

Ignores small intraday market noise to prevent fee-churning and excessive order submission.

► Strict Cooldown Windows

Enforces a mandatory 60-minute cooldown post-execution to allow price discovery to settle.

► Emergency Shock Override

Extreme tail-risk shocks ($>5\%$ market drop or $VIX > 35$) bypass cooldowns to trigger instant defense.

TIER-2: CLOSED-LOOP REASSESSMENT ACTIONS

- **INCREASE**

Market volatility spikes; scales hedge ratio up to target to absorb incoming shock.

- **DECREASE (The Differentiator!)**

Market stabilizes; systematically unwinds protection to lock in option gains and avoid theta decay.

- **REPLACE / ROLL**

Hedge approaches expiration or delta shifts out of the target corridor; rolls to optimal strike/expiry.

- **REMOVE**

Risk returns to baseline normal; closes remaining protection to return portfolio to 100% growth.

- **MAINTAIN**

Hedge is performing within optimal bounds; no action taken, saving transaction friction.

LIFECYCLE PROOF: AEGIS is not an options-buying bot. It knows when to harvest gains and remove hedges, returning capital to growth.

Institutional Dashboard: 'Alabaster Spruce' UI

Swiss private banking elegance combined with real-time quantitative telemetry and auditable decision trails.

PORTFOLIO KPI TELEMETRY

Real-Time Equity & Risk Exposure

- Live Net Liquidation Value & Uninvested Cash
- Current Drawdown vs Historical High-Water Mark
- Hedge Ratio Gauge (Current vs Target)
- Hedged P&L vs Unhedged Benchmark Overlay

INTERACTIVE PAYOFF CHART

Recharts Visual Convexity Model

- Multi-curve visualization: Unhedged vs Hedged
- Downside floor vs Capped upside collar visualization
- Interactive spot price slider & breakeven marker
- Real-time contract payoff expiration modeling

LIVE GREEKS MATRIX

Second-Order Sensitivities

- Portfolio Delta (Δ): Net directional dollar exposure
- Gamma (Γ): Rate of delta acceleration
- Theta (Θ): Daily time decay dollar cost
- Vega (v): Volatility shift sensitivity

AUDITABLE DECISION TRAIL

Full Regulatory & Judge Transparency

- Chronological log of every agent reasoning step
- Side-by-side strategy score comparison table
- Risk Gate approval/modification audit checklist
- Direct links to Alpaca order IDs and fill timestamps

DESIGN SYSTEM: Built with React 18, TypeScript, Tailwind CSS, Playfair Display (Serif), and JetBrains Mono (non-jittering tabular financial figures).

The 8-Scene Golden Narrative Replay

Our deterministic test replay engine proves the complete autonomous cycle from market shock to de-hedging.

1. Baseline State

Healthy tech portfolio (\$100k), VIX low at 14.2, hedge ratio 0%, No-Hedge active.

2. Volatility Shock

Tech sell-off starts; portfolio drawdown reaches 3.8%, VIX jumps to 26.5.

3. Ingestion Phase

5 Ingestion agents gather telemetry; standardized HedgeContext assembled.

4. Strategy Arena

4 agents formulate hypotheses: Put, Put Spread, Collar, and No-Hedge.

5. Strategy Selection

Put Spread wins: optimal balance of downside protection, budget, and upside.

6. Risk Gate Pass

Risk checks Greeks, cash budget (\$1,250 cap), liquidity, and verifies buying power.

7. Alpaca Execution

Multi-leg order sent via Alpaca API; both legs fill atomically with logged slippage.

8. Market Recovery

Market rallies; drift detected; agent autonomously executes DECREASE to lock gains!

REPRODUCIBILITY: Run ``make demo-restore`` or ``python -m backend.demo_replay`` to deterministically replay the full 8-scene sequence in your local environment.

Production Technology Stack & Quantitative Models

A robust, type-safe system built with high-concurrency Python, LangGraph, and PostgreSQL.

SYSTEM COMPONENT	TECHNOLOGY STACK & IMPLEMENTATION DETAILS
Core Backend	Python 3.11+, FastAPI, Pydantic v2, Uvicorn, Alembic database migrations
Multi-Agent Orchestrator	LangGraph stateful workflow DAG, StateGraph checkpointer, resilient retries
Dual LLM Provider Layer	Featherless AI (Competition submission) & OpenRouter (Development) via OpenAI API
Quantitative Engine	Black-Scholes analytical pricer, numerical Greek solvers, Cornish-Fisher VaR, Payoffs
Brokerage & Market Data	Alpaca Trading API, Alpaca Options Chains, Multi-leg Orders, Alpaca MCP Tools
Database & Persistence	PostgreSQL (Neon serverless pooled & unpooled DSN) storing runs, orders, & states
Frontend Application	React 18, TypeScript, Tailwind CSS, Vite, Recharts, Lucide Icons
DevOps & Testing Suite	Docker containerization, Makefile targets, pytest (unit/integration/smoke), CI GitHub Actions

ENGINEERING RIGOR: 100% type-checked with mypy/ruff, 8-phase verified test suites, full OpenAPI 3.1 contract compliance.

Judging Criteria Alignment & Commercial Roadmap

How AEGIS hits every evaluation criterion and scales into a commercial private wealth product.

1. ALPACA INTEGRATION Deep Platform Utilization ✓ Live Paper Trading Brokerage Account ✓ Real-time Options Chain Discovery ✓ Native Multi-Leg Execution (Spreads/Collars) ✓ Full Alpaca MCP Tool Calling Protocol	2. P&L & RISK PERFORMANCE Quantitative Downside Cushioning ✓ Verified Drawdown Reduction during shocks ✓ Eliminates Theta Drag via Spread funding ✓ Maintains upside participation during recoveries ✓ Net P&L tracking vs unhedged benchmarks	3. CREATIVITY & NOVELTY Breakthrough Agentic Architecture ✓ First closed-loop adaptive de-hedging agent ✓ 4-Strategy autonomous competition arena ✓ Strict separation of LLM reasoning vs math ✓ Auditable multi-step decision trails	4. EXECUTION & RELIABILITY Institutional Engineering ✓ Full 8-phase test verification suite ✓ Deterministic demo replay script ✓ FT Luxury Editorial UI design system ✓ Self-contained Docker & Make workflows
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COMMERCIAL ROADMAP: SCALING FROM HACKATHON TO INSTITUTIONAL SAAS ▶ Phase 1: Multi-User Wealth Portals: OAuth2 login, encrypted Alpaca user key vaults, role-based portfolio access for RIAs. ▶ Phase 2: Advanced Derivative Overlays: Tier-B options strategies: VIX Call Calendar spreads, Ratio backspreads, synthetic long collars. ▶ Phase 3: Tax-Loss Harvesting Engine: Combining adaptive options hedging with intelligent tax-lot harvesting across taxable equity portfolios.
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